

# Weather and Climate Big Data

기상기후빅데이터

Hands-on Exercise 4: Time Series Modeling

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# Today's Topics

- Time Series Data
- Time Series Modeling Approaches
- Hands-on: Temperature Prediction

# What is time series?

- A sequence of data points collected or recorded at (ir)regular time intervals
- Values are ordered in time, and the order matters (this is different from typical regression!)
- Shows temporal dynamics such as trends, seasonality, and autocorrelation
- Examples: daily temperature, rainfall, stock prices, web traffic

# Components of Time Series

- Time series data often consists of several underlying patterns that affect its behavior over time:
  - **Trend:** Long-term upward or downward movement
  - **Seasonality:** Regular, repeating patterns at fixed intervals
  - **Noise:** Random, unpredictable fluctuations

# Components of Time Series

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  - **Trend:** Long-term upward or downward movement
  - **Seasonality:** Regular, repeating patterns at fixed intervals
  - **Noise:** Random, unpredictable fluctuations
  - **Autocorrelation:** Current values depend on past values
  - **Stationarity:** Statistical properties (mean, variance) stay constant over time

# Time Series Analysis

- Time series analysis involves analyzing data collected over time to understand patterns and make predictions.
  - **Forecasting:** Predicting future values based on past data
  - **Anomaly Detection:** Finding unexpected or unusual patterns
  - **Trend Analysis:** Identifying long-term increases or decreases
  - **Clustering:** Grouping similar time series patterns or regions

# From Patterns to Predictions

- We've seen what time series are and what makes them unique. how do we actually model it (\*data-driven)?
  - **Statistical models:** grounded in mathematical assumptions
  - **Machine learning models:** flexible, data-driven
  - **Deep learning models:** capable of capturing complex temporal dependencies
- Each has its strengths and use cases - let's walk through them one by one!



# Time Series Modeling

- **ARIMA (Autoregressive Integrated Moving Average)**
  - A classical statistical model used for time series forecasting
  - **AR (Auto-Regressive)**: current value depends on past values
  - **I (Integrated)**: differencing is used to remove trend and make the series stationary.
  - **MA (Moving Average)**: current value depends on past forecast errors.
  - Works well for stationary data with linear patterns and short-to medium-term forecasts.



# Time Series Modeling

- **Random Forest**

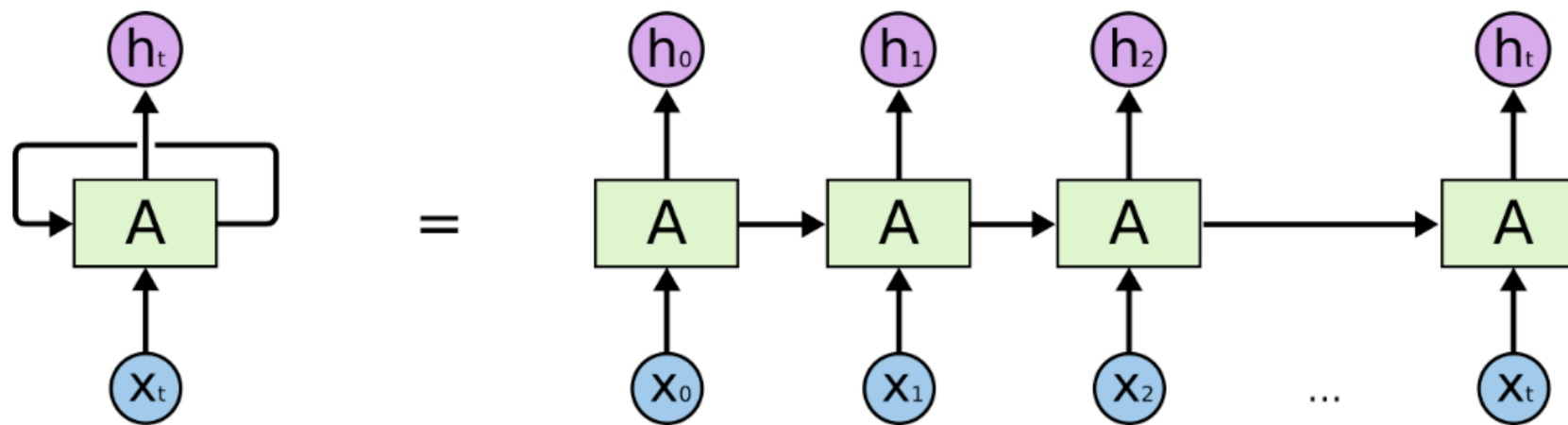
- An ensemble learning method that combines multiple decision trees for regression or classification.
- Handles nonlinear relationships and interactions between variables well.
- Requires feature engineering (e.g. lag values).
- Robust to overfitting and works well with missing or noisy data.

# Temperature prediction

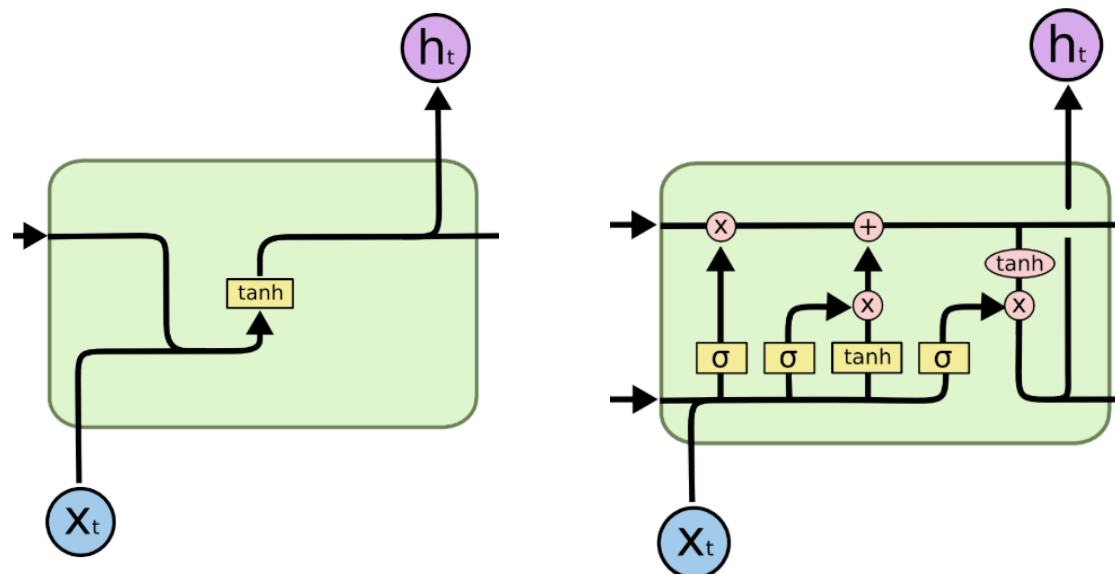
- **LSTM (Long Short-Term Memory)**
  - A type of recurrent neural network (RNN) designed to learn long-term dependencies in sequential data.
  - Uses memory cells and gates (input, forget, output) to retain and control information over time.
  - Excellent for modeling complex, nonlinear, and long-range patterns in time series.
  - Ideal for high-volume, multivariate time series data and long-term forecasting.

<https://research.google/blog/a-flood-forecasting-ai-model-trained-and-evaluated-globally/>

# Time Series Modeling



An unrolled recurrent neural network.



<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

# Time Series Modeling

- **PatchTST (Patch-based Time Series Transformer)**
  - A Transformer-based model specifically designed for time series forecasting
  - *Uses patching (like in computer vision)* to divide time series into segments, capturing local patterns more effectively
  - Eliminates the need for RNNs or positional encoding by leveraging patch structure
  - Achieves strong performance in long sequence forecasting, often outperforming LSTM and Informer (another Transformer-based model, but more complex)